Paleomagnetism of diabase dikes from the Bonavista Bay area of northeastern central Newfoundland

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Paleomagnetic results are reported for a swarm of diabase dikes from the northeastern Gander Zone of the Newfoundland Appalachians. Sixty-eight cores were collected from 19 dikes cutting the various granites and migmatites from the Bonavista Bay area of Newfoundland. From the earlier structural and Rb/Sr age data available, one can infer an age for the dike swarm of between 400 and 330 Ma. Paleomagnetic studies including alternating field and thermal demagnetizations reveal that the swarm can be divided into two groups: group 1 has a characteristic magnetization of $D = 296.6°$, $I = 82.2°$ with $\alpha_{95} = 5.8°$ for $N = 14$ dikes with the corresponding pole position at 53.9°N, 77.4°W (δp, $\delta m = 11.0°$, 11.3°; group 2 has a characteristic magnetization of $D = 157.6°$, $I = 53.9°$ with $\alpha_{95} = 9.0°$ for $N = 5$ dikes with the corresponding pole position at 3.9°S, 35.5°W (δp, $\delta m = 8.8°$, 12.6°). One dike's magnetization was found to be reversed relative to the other group 2 dikes. Various magnetic experiments (low- and high-field hysteresis, Curie temperatures, susceptibility variation with temperature, coercivity of remanence, Lowrie–Fuller test) conducted to study the bulk magnetic and remanent magnetic properties of the dike material suggest that their behavior is consistent with multidomain pure magnetite being the dominant magnetic mineral. The group 1 pole position lies close to Newfoundland and is similar to other poles of equivalent age for intrusives from New Brunswick and Newfoundland. Rock units from the southeastern margin of the Iapetus Ocean thus seem to be yielding consistent "high-latitude" poles, a factor that needs to be taken into account in any discussion of the tectonic evolution of the area.

On rapporte les résultats d'une étude paléomagnétique d'un réseau de dykes de diabase de la zone nord-est de Gander dans les Appalaches de Terre-Neuve. En tout, 68 carottes furent prélevées sur 19 dykes pénétrant divers granites et migmatites de la région de la baie Bonavista de Terre-Neuve. Les renseignements sur la structure et les datations au Rb/Sr disponibles permettent d'évaluer l'âge du réseau de dykes entre 400 et 330 Ma. Les études paléomagnétiques incluant la désaimantation par champ alternatif et par traitement thermique permettent de séparer ce réseau en deux groupes: le groupe 1 dont l'aimantation est caractérisée par $D = 296{,}6°$, $I = 82{,}2°$, avec $\alpha_{95} = 5{,}8°$ pour $N = 14$ dykes et dont la position correspondante du pôle est 53,9°N, 77,4°O (δp, $\delta m = 11{,}0°$, 11,3°); le groupe 2 dont l'aimantation est caractérisée par $D = 157{,}6°$, $I = 53{,}9°$ avec $\alpha_{95} = 9{,}0°$ pour $N = 5$ dykes et dont la position correspondante du pôle est 3,9°S, 35,5°O (δp, $\delta m = 8{,}8°$, 12,6°). On observe que l'aimantation de ce réseau de dykes est inversée par rapport aux dykes du deuxième groupe. Plusieurs mesures magnétiques (hystérèse à faible et grand champ), points Curie, variation de la susceptibilité avec la température, force coercitive de rémanence, essai Lowrie–Fuller) ont été effectuées pour étudier les propriétés d'aimantation apparente et rémanente des matériaux constituant les dykes, et elles indiquent que leur comportement est conforme avec la présence d'une magnétite pure à multi-domaines comme principal mineral magnétique. La position du pôle pour le groupe 1 est localisée près de Terre-Neuve et coïncide avec la position des autres pôles antérieurement observés pour les roches intrusives du même âge au Nouveau-Brunswick et à Terre-Neuve. Les unités lithologiques de la marge sud-est de l'océan Iapetus semblent donc présenter des pôles concordants avec une "latitude élevée," un facteur qui doit être considéré dans toute discussion portant sur l'évolution tectonique de cette région.

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Introduction: Geology of the area and objectives of the investigation

The distinct features of the geology of the island of Newfoundland are the western platform and the Avalon Peninsula, comprising Precambrian basement, joined together by Central Mobile Belt oceanic material of probable Paleozoic age. During the past few years, interpretations of the geology of the island has undergone rapid changes based on Wilson's (1966) hypothesis of a proto-Atlantic Ocean and the current ideas on plate tectonics. The revised interpretations are efficiently reviewed in a series of articles by Williams *et al.* (1972, 1974). They fitted the geology of the island into various tectono-stratigraphic zones, each zone having a characteristic tectonic, stratigraphic, and metamorphic signature. These zones, labelled A–H from west to east (Fig. 1), are extrapolated (Williams 1978) to cover the full Appalachians to their southern extremity in the United States.

Of the various tectono-stratigraphic zones, the Gander Zone is quite complex geologically and is characterized by highly metamorphosed and poly-deformed rocks. Some of the prevailing views on the geological evolution of the area are (i) the Gander Zone was the ancient continental margin of the Euro-African continent prior to closing of the proto-Atlantic

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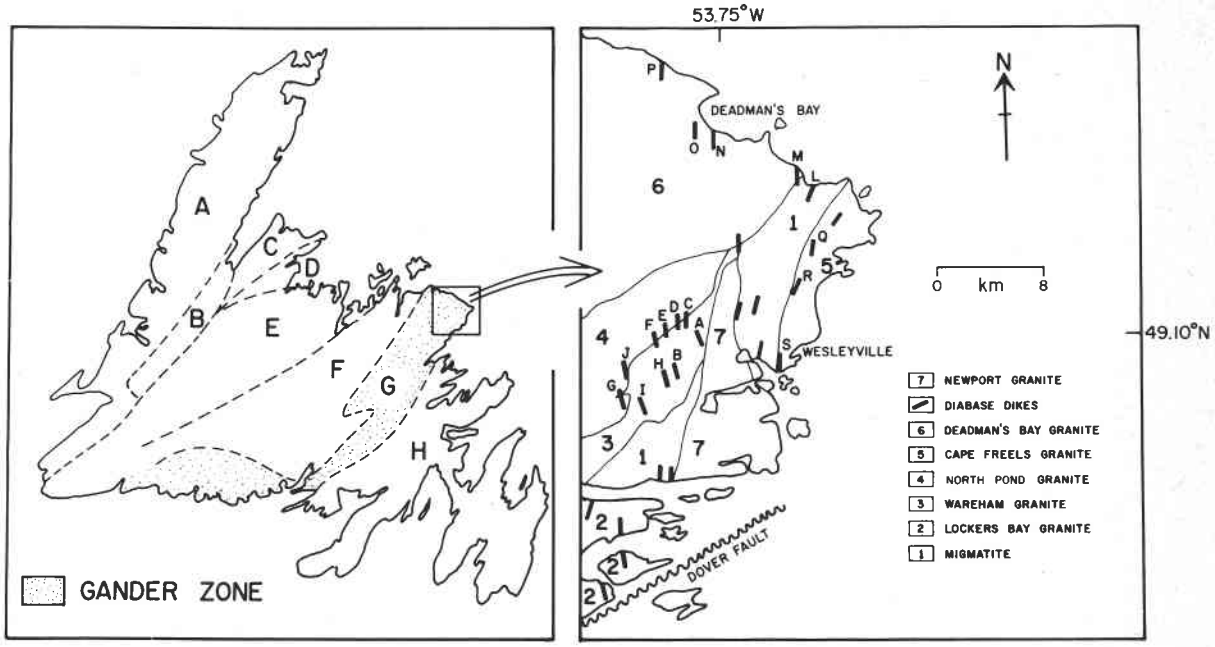


FIG. 1. Generalized geological sketch map of northeastern portion of the Gander Zone showing paleomagnetic sampling localities of the Bonavista Bay dikes (A–S). Also shown are the boundaries of various granite plutons in the area and of the migmatites. The numbers 1–7 are from oldest to the youngest rock unit in the area. The tectono-stratigraphic zoning in Newfoundland (after Williams *et al.* 1972) is shown to the left.

(Williams 1964); and (ii) Avalon Peninsula was a microcontinent with Gander Zone forming its western margin, the western margin of ancient Euro-Africa being somewhere east of the Avalon Peninsula (Schenk 1971). Geological interpretation of this area is further complicated by the presence of the Dover Fault (Fig. 1), which forms a distinct boundary between the poly-deformed Gander Zone rocks and the relatively undeformed Precambrian Love Cove Group of the Avalon Peninsula (Blackwood and Kennedy 1975). Gander Zone separates the relatively undeformed Ordovician–Silurian sedimentary and volcanic rocks of the Botwood Zone to the west and also the relatively undeformed Hadrynian–Ordovician sedimentary and volcanic rocks of the Avalon Zone. In contrast the Gander Zone rocks are highly metamorphosed and poly-deformed. In the area of interest, in the northeast Gander Zone, the country rocks are migmatites and gneisses of probable sedimentary origin (the Hare Bay gneiss complex and the Bonavista Bay gneiss complex of Kennedy (1975); Blackwood and Kennedy (1975)). These gneisses are overlain by dominantly metasedimentary rocks of the Gander Group (Kennedy 1975). The original mapping of the area was done by Jenness (1963).

A distinct feature of the northeast part of Gander Zone is the presence of numerous granite plutons. Although earlier structural evidence (Kennedy and McGonigal 1972; Blackwood and Kennedy 1975; Kennedy 1975) suggested a pre-mid-Ordovician age for the intrusion of these granites, more recent structural studies (Jayasinghe and Berger 1976; Pajari and Currie 1978; Hanmer 1981) and radiometric ages (Bell and Blenkinsop 1977; Bell *et al.* 1977, 1979; Dallmeyer *et al.* 1981) showed a mid-Devonian age for the granites. Jayasinghe and Berger (1976) drew attention, in particular, to the presence of a suite of diabase dikes that intrude most of the granites of the area and in turn are intruded by the Newport granite, which yielded the youngest age by Rb–Sr methods (330 Ma). Thus the diabase dike intrusion probably took place (surprisingly) in the middle of sporadic or continuous granite intrusion confined to a very narrow time period of 420–330 Ma. If the radiometric ages of the granites indicate the ages of intrusion of the granites, the time of intrusion of the basic dikes can be placed with good certainty as between 400 Ma (age of Cape Freels granite) and 330 Ma (age of the Newport granite). This report on the paleomagnetism of diabase dikes forms part of a comprehensive study of the Gander Zone rocks that was undertaken with the hope that it could lead to a choice between some of the presently held views on the tectonic evolution of this area.

Figure 1 shows the paleomagnetic sampling sites

TABLE 1. Paleomagnetic results of individual dikes in the present study

	NRM					After demagnetization[b]					
Dike	N	$\bar{D}$ (°)	$\bar{I}$ (°)	α_{95} (°)	κ	$\bar{D}$ (°)	$\bar{I}$ (°)	α_{95} (°)	κ	R	δ_w
Group 1											
A	4	4.9	75.2	16.8	30.7	344.0	79.5	8.5	117	3.9744	6.5
B	4	330.3	85.4	13.2	49.6	176.3	82.2	3.3	794	3.9962	2.5
C	4	293.6	73.9	22.5	17.7	300.2	82.2	8.5	117	3.9744	6.5
D	3	303.3	75.4	19.5	40.9	306.2	89.6	8.2	226	2.9912	4.4
E	3	357.6	63.8	17.8	20.5	290.5	87.8	14.4	74.5	2.9732	7.7
F	3	194.5	81.4	19.2	17.8	223.6	77.2	12.1	105	2.9810	6.4
G	3	345.8	74.6	15.5	64.5	309.9	84.8	10.8	131	2.9847	5.8
H	3	324.5	83.0	28.9	19.2	227.1	75.4	13.6	82.8	2.9758	7.3
J	4	324.6	82.2	12.0	59.0	356.2	82.5	6.2	218	3.9862	4.8
K	3	308.8	78.3	41.6	9.8	310.1	78.9	8.8	198	2.9899	4.7
L	4	311.0	70.8	17.0	30.0	331.2	81.4	7.9	135	3.9778	6.0
P	4	338.0	71.0	17.8	27.6	338.0	78.0	8.7	112	3.9732	6.6
Q	3	300.5	81.1	12.8	93.0	271.0	70.0	12.6	96.2	2.9792	6.8
R	4	306.9	61.3	17.9	27.3	314.0	55.0	12.1	58.7	3.9489	9.2
Group 2											
I	4	120.6	87.7	18.5	25.7	140.5	54.5	10.9	72.3	3.9585	8.3
M	4	342.3	64.2	25.5	13.9	340.0	−65.0	14.2	42.6	3.9296	10.8
N	4	104.7	85.0	50.6	4.3	160.0	55.0	15.1	37.8	3.9206	11.4
O	3	135.5	81.8	22.2	31.8	160.5	48.0	11.6	113	2.9823	6.2
S	4	232.4	86.2	31.7	9.3	165.5	45.5	17.5	28.5	3.8947	13.2

	Pole position												
Dikes	Lat.	Long.	δ_p (°)	δ_m (°)	A_{95} (°)	Paleolatitude	N	$\bar{D}$ (°)	$\bar{I}$ (°)	α_{95} (°)	κ	R	δ_b (°)
Group 1 (BB1)	53.9°N	77.4°W	11.0	11.3	9.7	74.5°N	14	296.60	82.2	5.8	47.9	13.7284	11.3
Group 2 (BB2)	3.9°S	35.5°W	8.8	12.6	10.6	34.5°N	5	157.60	53.9	9.0	72.7	4.9450	8.5

NOTES: For basis of classification of dikes, see text; N = number of cores from each dike; $\bar{D}$ = mean declination; $\bar{I}$ = mean inclination; α_{95} = radius of circle of 95% confidence; κ = estimate of Fisher (1953) precision parameter; R = length of resultant of N vectors; δ_w = within dike precision; δ_b = between dike precision; δ_p, δ_m are the semi-axes of oval of 95% confidence; A_{95} = radius of circle of 95% confidence around the pole.

[a]Direction of dike M is inverted in calculating the group 2 mean.

[b]Treatment step used is not given here as for any dike both AF and thermal demagnetization procedures were used and different samples from any given dike have different fields or temperatures taken as optimum.

from the basic dikes. The dikes strike north–south and are subvertical. The thicknesses range from less than 1 m to about 5 m. The dikes are exposed in an area of approximately 800 km^2 in the northeast section of the Gander Zone. Jayasinghe (1978) described the petrography and geochemistry of dikes mostly around Wesleyville and speculated on the possibility that these dikes mark a localized tensional event in the Devonian, which was a precursor to the major continental breakup that occurred later. The sampled dikes are identified by letters A–S (Fig. 1). The dike material seems to be extremely fresh. At least three oriented cores were drilled from each dike (Table 1). The orientation of cores was by both solar and magnetic compasses. A total of 342 individual samples were cut from these cores for the paleomagnetic investigations.

Paleomagnetism

The following magnetic parameters were measured or calculated for the samples. The remanent magnetization, both natural and after partial demagnetization, the initial susceptibility, Koenigsberger ratios, and the presence or absence of Rayleigh loops. A Schonstedt SSM-1 spinner magnetometer was used for measuring the remanent magnetism, a susceptibility bridge described by Patzold (1972) for measuring the intial susceptibility, and a Rayleigh loop tracer described by Likhite and Radhakrishnamurty (1966) for checking low field (1 mT) hysteresis called the Rayleigh loops. The values of Koenigsberger ratios are typically between 0.1 and 0.4, which seem to be lower than normal values expected for basic dikes. The values of susceptibility fall in a narrow range of 0.90×10^{-2} to 3.0×10^{-2} cgs

units. The natural remanent magnetization (NRM) moments are clustered around 10^{-5} A m^2. The magnetizations of a minimum of two samples per core were measured. They were combined to obtain a mean magnetization for the core. The magnetizations of all the cores from a dike were combined to obtain mean direction of magnetization for each dike. These values at the NRM level are plotted in Fig. 2 and are listed in Table 1. The magnetization is uniform within each dike and the magnetizations between dikes also agree fairly well (Fig. 2; Table 1). Steep positive inclinations were noticed for all the specimen directions. Alternating field (AF) and thermal demagnetizations of the specimens were carried out to isolate the characteristic magnetization for each dike. All specimens were treated successively in either 5 mT steps or at temperatures of 100, 200, 300, 400, 500, 520, 540, 560, 600, 620, 640, and 680°C. All demagnetizations were carried out in ambient fields of not more than 50 nT during AF demagnetization and ~25 nT during thermal demagnetization.

Most of the specimens have median destructive fields below 10 mT, suggesting that the specimen's magnetism resides in predominantly low coercive force, multidomain grains. The dikes can be divided into two groups based on their response to demagnetization.

In group 1 dikes (which are a majority), the directions moved to steeper inclinations, mostly with northwest declinations. As may be seen in Fig. 3, reasonably stable end points were noticed during AF treatment. Intensity values fell to about 10% of their original value in fields of 10 mT. A similar behavior was noticed during thermal demagnetization, with intensities falling less rapidly than in the AF case, except for the highest temperature range (Fig. 4). It is likely from the results of the demagnetization experiments, particularly from the persistence of the steep inclinations at high fields and at high temperatures, that the closeness of the NRM direction to that of the present Earth's field is coincidental and that this group of dikes has a characteristic magnetization with steep inclinations. Each specimen from this group of dikes was demagnetized in all the steps at successively increasing temperatures or fields. There is improvement in grouping on demagnetization within dikes, as may be noticed in Table 1.

In group 2 dikes, the north-seeking directions moved systematically to southeasterly declinations with moderate to steep inclinations. Stable end points were obtained during both AF and thermal demagnetizations (Figs. 5, 6). There is a general improvement in grouping both within and between dikes. Directions for all specimens from one dike (site M) changed systematically on demagnetization from steep positive inclinations to steep negative inclinations. The final dike mean direction for this site is 180° to the directions of the other group 2 dikes. This is probably a dike that is reversely magnetized with respect to the group 2 dikes.

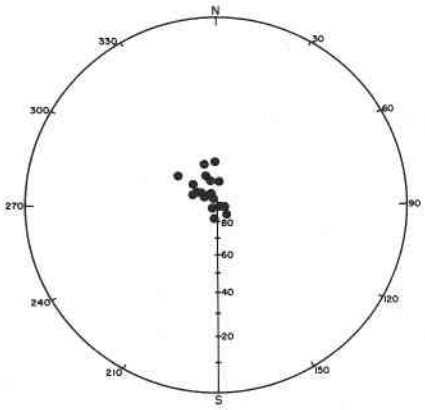


FIG. 2. NRM mean directions of the Bonavista Bay dikes. All are positive inclinations, plotted on Wulff's stereographic projection.

On thermal demagnetization beyond 600°C, specimen directions became random and intensity of magnetization fluctuated wildly. It is concluded that there is no recoverable paleomagnetic signal in the specimens beyond the 600°C step, suggesting that magnetite is probably the magnetic mineral in the samples. The final directions of the specimens were selected after either AF or thermal demagnetization, based solely on stability of the end point. For most of the specimens, this is either the 40 mT or the 540°C step. The final demagnetized directions of the specimens were combined at the dike level as before; the dike mean directions are listed in Table 1 and are shown in Fig. 7.

Tests on stability of magnetization

Baked contact test

Response of specimens to demagnetization is characterized by an almost exponential drop of remanent intensities. Further, the group 1 directions are particularly close to the direction of the present Earth's field. These factors may suggest a possible soft magnetization for the dikes. In order to check whether the magnetizations obtained for the dikes are indeed primary, the host rock was sampled for baked contact tests for the two groups of dikes. Dike F (group 1) intrudes the Wareham granite and dike S (group 2) intrudes the Cape Freels granite. Samples were systematically drilled at close spacings starting 3 cm away from the dike and extending to about 5 m. The directions of magnetization of the host rock specimens are plotted in Fig. 8 at NRM level and after demagnetization. Figure 8 *a* and *b* for Wareham granite shows a characteristic magnetization with steep positive inclination and northwest declination obtained after

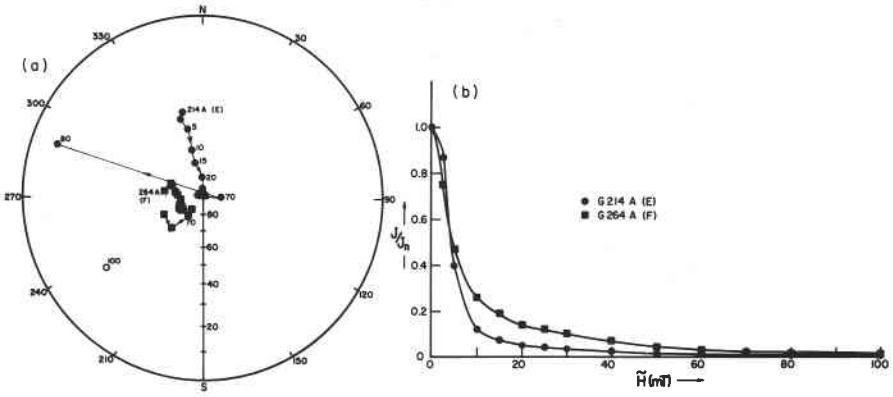


FIG. 3. Typical response of specimens from group 1 dikes to AF demagnetization. (*a*) Change of directions of G214A from dike E and of G264A from dike F with demagnetization, plotted on a Wulff's net. Solid (open) symbols indicate positive (negative) inclinations. Numbers next to the symbols indicate the values of the corresponding peak alternating field in mT to which the specimen was exposed. (*b*) Corresponding variation in intensity. For basis for classification of dikes into groups, see text.

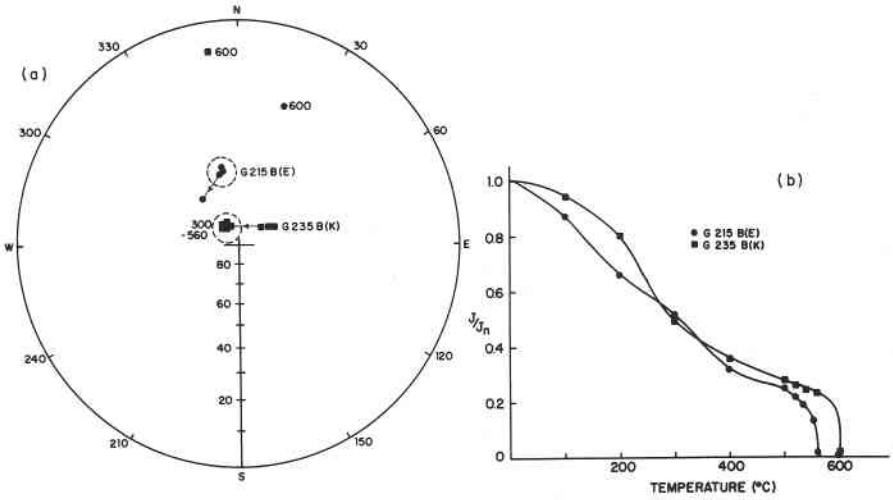


FIG. 4. Response of specimens from group 1 dikes to thermal demagnetization. (*a*) Change of directions of G215B from dike E and G235B from dike K with temperature, plotted on Wulff's net. All are positive inclinations. Dotted circle around a set of symbols indicates that the specimen direction remained within the circle for the temperature interval (degrees Celsius) indicated. (*b*) Corresponding variation in intensity.

15 mT AF demagnetization. Thermal demagnetization of specimens from this granite showed scattered results. Figure 8 *c* and *d* for the Cape Freels granite intruded by a group 2 dike shows a characteristic magnetization after thermal demagnetization (Fig. 8*d*) with northerly declination and a shallow negative inclination, a direction that is in no way similar to that of the dike. AF demagnetization of specimens of this granite yielded a direction similar to that after thermal demagnetization. Thus, there is no doubt that this direction is characteristic for this granite. No systematic change in direction of magnetization of the specimens was noticed with distance away from the dike for either granite. The contact test was thus inconclusive for the group 1 dikes as both the dike and the host rock have similar characteristic magnetizations. For the group 2 dikes, the fact that the host rock has an entirely different characteristic magnetization and further that the magnetization of the host rock does not seem to be effected by the dike intrusion suggests that the group 2 magnetization may not be primary.

Storage test

A belated cursory storage test was attempted for the dike specimens, particularly group 1 dikes, to determine if the source of their directions is the present Earth's field. Thirty specimens (15 fresh ones and 15 that were previously AF demagnetized to various levels ranging

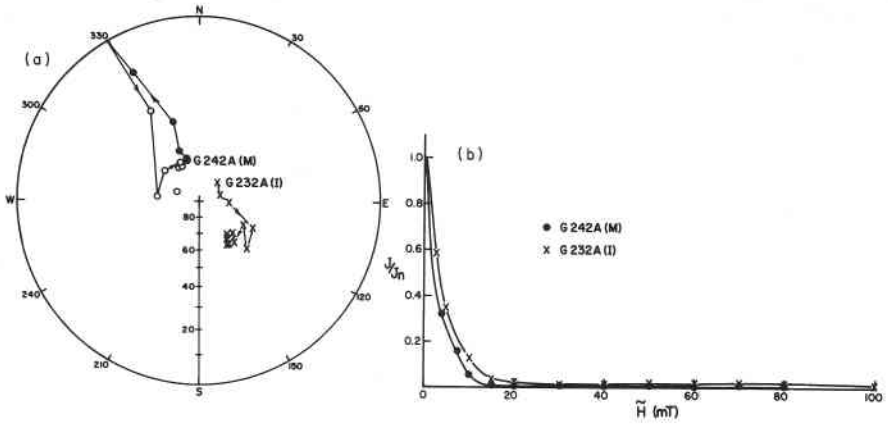


FIG. 5. Response of specimens from group 2 dikes to AF demagnetization. (*a*) Change of directions of G232A from dike I and G242A from dike M with demagnetization. (*b*) Corresponding variation in intensity. See caption for Fig. 4 for explanation of symbols and numbers.

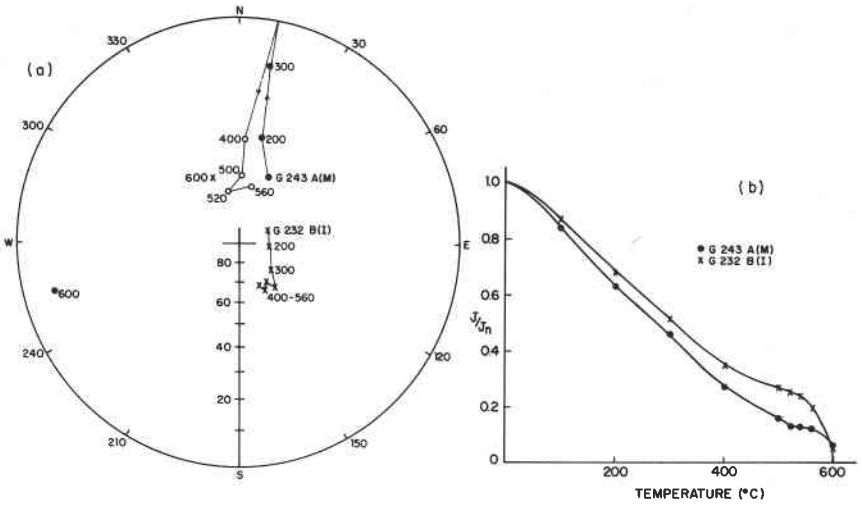


FIG. 6. Response of specimens from group 2 dikes to stepwise thermal demagnetization. (*a*) Change of directions of G232B from dike I and G243A from dike M with temperature, plotted on a Wulff's net. Solid (open) symbols indicate positive (negative) inclinations. Numbers next to the symbols indicate values of the corresponding temperatures to which the specimen was heated. (*b*) Corresponding variation in intensity.

from 15 to 60 mT) were measured for their NRM and then placed in the present Earth's field (PEF). They were placed in such a way that the PEF was perpendicular to the remanence of the specimen; its effect was to decrease the remanence inclination without any effect on declination. The specimen magnetization was remeasured after an interval of 6 weeks. It was noticed that in the fresh specimens there was no change in the inclination (~1°) whereas in the previously demagnetized specimens the inclinations shifted by as much as 40°. The declination hardly changed during this experiment. It is only to be expected that there would be a considerable change in direction of the previously demagnetized specimens as most of the magnetic grains in these specimens are in a randomized state due to demagnetization, and any biasing field would align these randomized magnetizations into its specific direction. It was further noticed that when these specimens were left in the laboratory without any specific orientation they recovered their original steep inclinations in a matter of hours. Although the storage test conducted here is not classical in the traditional sense, it may still be concluded from the behavior of the fresh specimens (constancy of directions) that it is probable their remanence is not caused by the PEF.

Anisotropy test

It may be possible that, due to nature of the rock fabric, the directions of magnetization of the dikes may be systematically deflected away from the ambient field

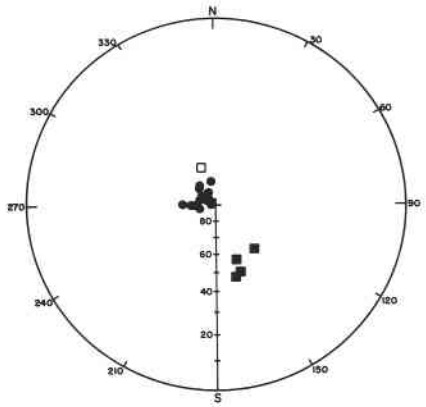


FIG. 7. Mean characteristic magnetization of the Bonavista Bay dikes obtained after demagnetization, plotted on a Wulff's net. Solid (open) symbols indicate positive (negative) inclinations.

direction. An anisotropy test, following procedures similar to those of Irving and Park (1973), was conducted for specimens from dikes H and O belonging to groups 1 and 2, respectively. Isothermal remanent magnetization (IRM) was induced in the specimens in the direction of their characteristic magnetization. The IRM was measured and then the specimens were progressively demagnetized by alternating fields; Fisher's statistics were calculated at each step. It was found that there was no significant bias of the IRM directions (mean) away from the direction of the applied field for either dike, the two directions agreeing to within 1.5° (α_{95} for grouping of the demagnetized IRM directions is ~3°). Thus it was concluded that the anisotropy of the rock unit, if any, is not sufficient to deflect the directions of characteristic magnetization away from that of the ambient field direction at the time the dikes were magnetized.

Magnetic experiments conducted to discern the nature of the magnetic minerals and their domain state, particularly of the remanence carriers, included the determination of the Curie points, study of the variation of susceptibility with temperature, determination of hysteresis loops (at room temperature and at liquid nitrogen temperatures) in high fields, determination of low-field hysteresis (Rayleigh loops), determination of the coercivity of remanence, and comparison of AF demagnetization characteristics of the isothermal and natural remanences. None of the specimens exhibited Rayleigh loops, indicating that the magnetic grains in the specimens are not in the effective superparamagnetic size range.

Susceptibility–temperature curves

Variation of susceptibility with temperature was studied from liquid nitrogen temperature to 700°C using apparatus described by Patzold (1972). A representative curve (Fig. 9*c*), identical for the two groups of dikes, shows a peak in normalized susceptibility at −150°C, a steady value up to ~560°C, a Hopkinson peak at 570°C, and a sharp drop to zero at 575°C. This type of behavior is interpreted as due to pure magnetite grains whose bulk magnetic behavior is multidomain (Radhakrishnamurty *et al.* 1978).

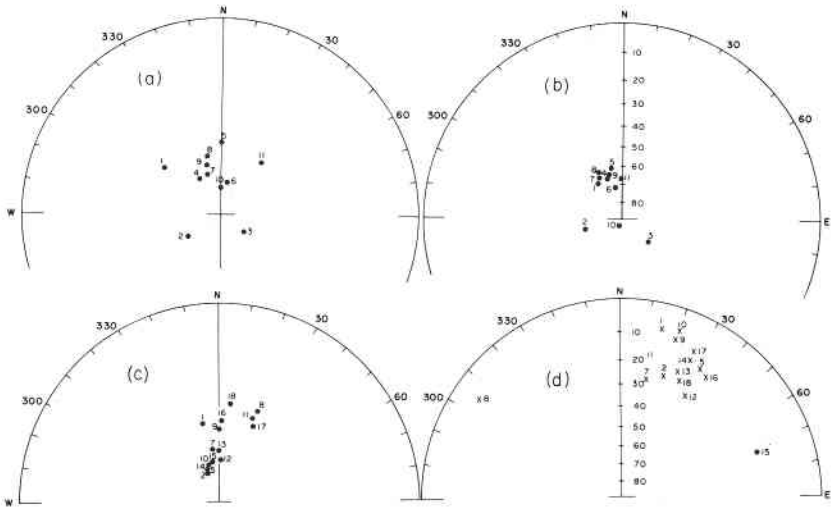


FIG. 8. Results of the baked contact test. Specimen magnetization directions of the host rock at the NRM level (*a*, *c*) and after demagnetization (*b*, *d*). (*a*), (*b*) For the Wareham granite intruded by a group 1 dike. Numbers next to the symbols are specimen numbers and increase with distance from the dike. The closest, No. 1, is 5.0 cm, and the farthest, No. 11, is 2.0 m away from the edge of the dike. (*c*), (*d*) For the Cape Freels granite intruded by a group 2 dike. The closest, No. 1, is 5.0 cm, and the farthest, No. 18, is 5.5 m away from edge of the dike.

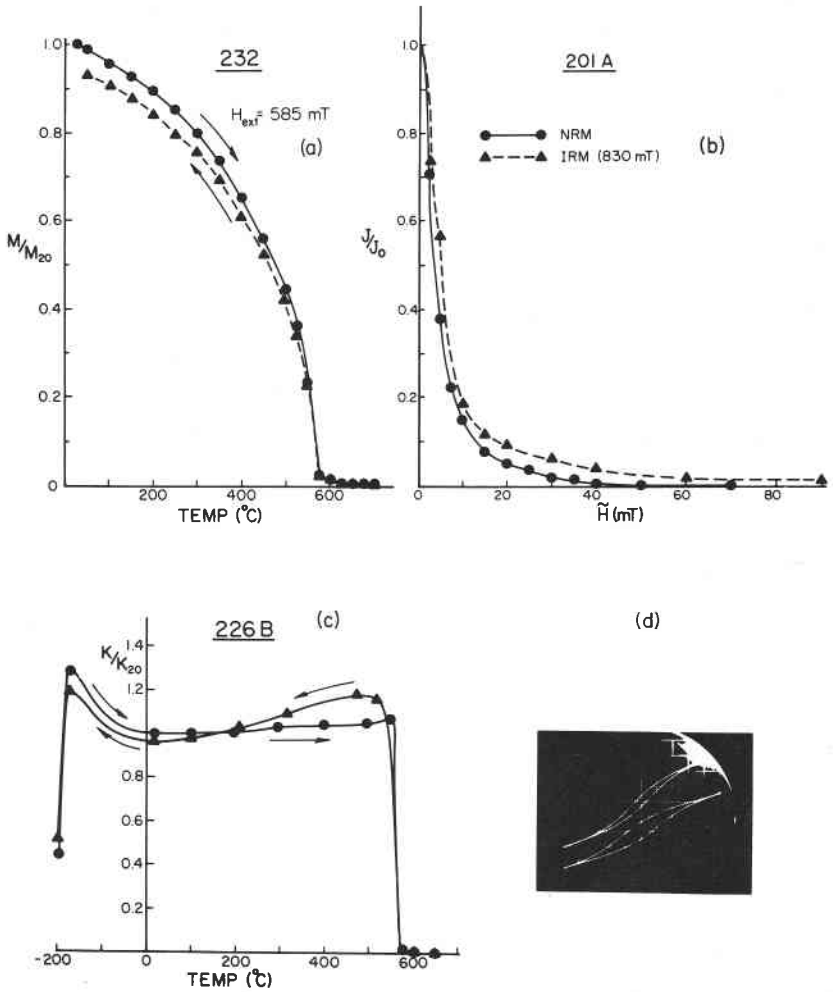


FIG. 9. (*a*) Normalized magnetic moment versus temperature in a field of 585 mT for G232 (heating and cooling). (*b*) Response of NRM and IRM for G201A to alternating field demagnetization (Lowrie–Fuller test). (*c*) Normalized susceptibility versus temperature curve for G226B in a field of 0.031 mT (heating and cooling). (*d*) Photograph of hysteresis loops obtained in a field of 120 mT peak field for specimen G223 at room temperature (upper trace) and at −196°C (bottom trace).

J_s–T *curves*

J_s–T curves were obtained in a field of 585 mT using an apparatus similar to the one described by Deutsch *et al.* (1971). Curves are identical for the two groups of dikes, and a typical one is shown in Fig. 9*a*. A smooth drop in magnetic moment at 575°C suggests that magnetite is the magnetic mineral in these rocks. Similarity between heating and cooling curves (Fig. 9 *a* and *c*) suggests that very little transformation of the magnetic mineral took place during heating.

Hysteresis loops

Hysteresis loops were obtained using apparatus similar to the one described by Likhite *et al.* (1965); representative curves are shown in Fig. 9*d*, both at −196°C (bottom trace) and at room temperature (top trace). Loops of this type, with an increase in coercivity at −196°C and a J_r/J_s ratio of around 0.2 were interpreted earlier as due to magnetic grains in multidomain state (Radhakrishnamurty *et al.* 1978).

Lowrie–Fuller test

In a paleomagnetic study, it is of interest to discover the domain state of the *remanence* carriers. Lowrie and Fuller (1971) suggested a simple procedure whereby one can determine the nature of the remanence carrier by studying the response to AF demagnetization of the natural remanence and the isothermal remanence imparted to a specimen. Figure 9*b* shows that the isothermal remanence is more difficult to demagnetize than the natural remanence, indicating that the remanence carriers are probably in multidomain form.

TABLE 2. Magnetic properties and domain classification of samples

Sample	J_r/J_s	H_{cr}/H_c	Lowrie–Fuller test	K–T curves	J_s–T curves
201	0.20	2.6	MD	MD	Reversible (T_C = 570°C)
225	0.15	2.2	MD	MD	Reversible (T_C = 570°C)
262	0.17	1.8	MD	MD	Reversible (T_C = 570°C)

NOTE: MD = multidomain.

This result is typical of specimens from both groups of dikes and is one of the rare examples where the Lowrie–Fuller test indicates a multidomain remanence carrier. The coercivity of remanence was determined by applying back field to saturation isothermal remanence imparted to the specimen. A value of 20 mT for H_{cr} was obtained for specimens from both groups of dikes, indicating that magnetite is the magnetic mineral.

Results of the various magnetic experiments are presented in Table 2. They suggest that multidomain magnetite is responsible for the magnetic behavior in these specimens. It is hard to visualize that multidomain grains can carry stable remanence with coercivities greater than 40 mT. It should be noted that stable end points are achieved (in the case of AF demagnetizations) at fields well *above* the sharp drop in the demagnetization curves where J/J_0 is very small. It is possible that the stable remanence for these specimens resides in a very small single-domain component that is completely masked by the dominant multidomain grains both in granulometric and Lowrie–Fuller type experiments.

The mean directions of individual dikes after demagnetization are plotted in Fig. 7 and also listed in Table 1. The dike means separate into two distinct groups. Group 1 (hereafter called BB1) dikes are characterized by steep positive inclinations and northwest declinations. Group 2 (hereafter called BB2) dikes have a characteristic magnetization with southeast declinations and moderate to steep positive inclinations. Improvement in grouping, stable end points, shift of directions to steeper inclinations for group 1 dikes, and the presence of a reversely magnetized dike within the group 2 dikes allow one to conclude that characteristic magnetizations for the two groups of dikes were isolated. The corresponding paleomagnetic poles are listed in Table 1.

Discussion of results

The age of intrusion of the dikes is known to be between 400 and 330 Ma from earlier structural evidence and Rb/Sr ages on the granites. P. Reynolds (personal communication, 1982) recently obtained an $^{40}Ar/^{39}Ar$ age of 375 ± 10 Ma for a dike from the southern end of the swarm, consistent with the other age relationships in the area. It should be noted that the dike for which the $^{40}Ar/^{39}Ar$ age was obtained is not among the ones for which paleomagnetic studies are reported here. Thus, although there is some age uncertainty for the magnetizations obtained in this study, it may be said that the age of intrusion of the dikes is likely to be around 375 Ma. All the available pole positions from the northern Appalachians for the Siluro-Devonian age are plotted in Fig. 10.

The two pole positions reported in this paper, BB1 and BB2, are "anomalous" in that they do not conform to the established pole positions for this age relative to North America. Hence the discussion in this paper is confined to these and other anomalous pole positions from the northern Appalachians. The two distinct pole positions (BB1 and BB2) obtained for the dikes may or may not be representative for the 375 Ma period. Such a situation, where the same rock unit leads to two distinct magnetizations, is not uncommon. Roy *et al.* (1979) reported two characteristic magnetizations of the St. George pluton of New Brunswick. Lapointe (1979) reported a high-latitude pole and another at equatorial latitude for the Mount Peyton batholith of Newfoundland. In the case of the Bonavista Bay dikes, such distinct magnetizations may mean (i) tectonic movement of one part of the area relative to the other; or (ii) different magnetization ages because a zone of deformation affected one part of the area preferentially, thus remagnetizing rocks in that part of the area. No amount of tectonic rotation can explain the BB1 magnetization. Hanmer (1981) showed that the eastern part of the area had preferentially undergone a shear strain. It is likely that the dikes in the eastern portion of the area also underwent this strain, with the resulting BB1 magnetization. This is further supported by the observation that the dikes yielding the BB1 pole are in general to the east of the dikes yielding the BB2 paleopole.

The presence of other high-latitude paleomagnetic poles (Fig. 10), all in the same general region, supports such a possibility and may help demarcate the shear zone

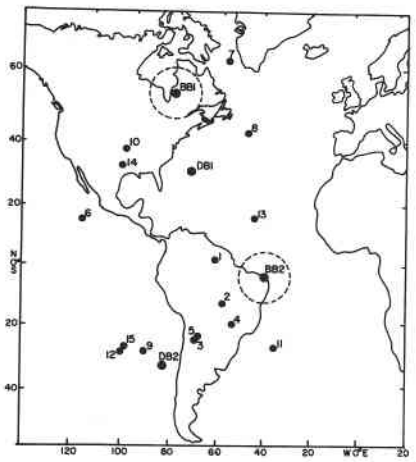


FIG. 10. Siluro-Devonian paleopoles from northern Appalachians. (1) Mascarene intrusives;[a] (2) Botwood rhyolite;[b] (3) Botwood sandstone;[b] (4) Hershey Formation;[c] (5) Eastport Formation;[c] (6) Mt. Peyton diorite;[b] (7) Mt. Peyton granite;[b] (8) St. Stephen pluton;[a] (9) St. George pluton;[a] (10) St. George pluton;[a] (11) Clam Bank Group;[d] (12) Traveler felsite;[e] (13) Cape Breton granite;[f] (14) Cape Breton granite;[f] (15) Cape Breton granite;[f] (DB1) Deadman's Bay diabase dikes;[g] (DB2) Deadman's Bay diabase dikes;[g] (BB1) Bonavista Bay dikes; (BB2) Bonavista Bay dikes. [a]Roy *et al.* (1979); [b]Lapointe (1979); [c]Kent and Opdyke (1980); [d]Black (1964); [e]Spariosu and Kent (1980); [f]Rao *et al.* (1981); [g]Murthy (1983).

of Hanmer (1981). However, attention should be drawn at this stage to Kent and Opdyke (1980) who dismissed, in general, the magnetization responsible for high-latitude poles because of (i) insufficient resolution techniques and because (ii) magnetization of intrusives is not a reliable indicator of paleomagnetic record. However, Cisowski and Wasilewski (1980) and Murthy *et al.* (1981) showed that intrusive type rocks often carry stable remanences that are good indicators of paleomagnetic record. Further, all the rock units that yielded high-latitude poles in this region had been well demagnetized and the remanences well resolved. Lapointe *et al.* (1978) discussed the matter of high-latitude poles in general and outlined some possible reasons for their limited number. Relative to the region of the Appalachians, there are too many high-latitude poles to justify dismissing them as result of inadequate resolution. It is suggested here that the high-latitude paleomagnetic poles for this age group relative to the Appalachians have to be explained either in terms of (i) rapid geomagnetic field excursion or as (ii) a feature peculiar to the northern Appalachians, having an explanation similar to that suggested by Irving (1979). It is interesting to note that all the high-latitude poles from Newfoundland and New Brunswick are from the southeastern side of the proposed Iapetus Ocean. Although we do not have, at present, a tectonic model to explain this feature, consideration should be given to incorporating such poles in any model for the tectonic evolution of the area.

It should also be borne in mind, as Irving and McGlynn (1981) correctly point out, that it is not the present latitude of a pole but the paleolatitude and rotation that can be calculated from them that are physically and geologically important. The steeply inclined BB1 magnetization yields high paleolatitudes (74.5°, Table 1) that fit Africa – South America better than North America. The BB2 magnetization yields an intermediate paleolatitude (34.5°) the significance of which is unclear at the moment.

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